230514: metarhyolite, Junior Hill prospect

(Little Peters Formation, Gindalbie Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Leonora (SH 51-1), Weebo (3141)
MGA Zone 51, 310325E 6865442N

Sampled on 12 July 2018

This sample was collected from the 335.81 – 336.73 m depth interval of drillcore JHDD00014, drilled in 2008 by St Barbara Limited at their Junior Hill prospect (Mukherji, 2009). The drillhole is located on Weebo Station and northwest of the Teutonic mining centre, about 7.2 km west-southwest of Garden Well, 6.8 northwest of Junior Hill, and 4.3 km northeast of Clifford Bore.

Geological context

The unit sampled is a metarhyolite (Fig. 1) assigned to the Little Peters Formation of the 2694–2676 Ma Gindalbie Group (Wyche et al., 2016, and references within). The Little Peters Formation consists mainly of felsic metavolcanic and metavolcaniclastic rocks interbedded with metasandstone and shale (Stewart, 2004; Sapkota and De Paoli, 2016). The formation overlies the Kents Bore Basalt and is tectonically juxtaposed against mafic and ultramafic rocks of the Marshall Pool Subgroup and metasedimentary rocks of the Mount Leonora Formation (Wyche et al., 2016). The present sample was collected to determine the age of fragmental metarhyolite above black shale. A metadacite from about 230 m depth in this drillcore yielded an igneous crystallization age of 2650 ± 6 Ma (GSWA 230507, Wingate et al., 2022). Zircons from a felsic igneous rock about 5.7 km to the east-southeast in drillcore JHDD00009 yielded a SHRIMP U–Pb date of 2692 ± 7 Ma (sample LNN009794, Mukherji, 2009). A porphyritic metadacite of the underlying Teutonic Bore Formation, sampled about 13.7 km to the south-southeast, yielded an igneous crystallization age of 2692 ± 4 Ma (GSWA 112159, Nelson, 1995). Pidgeon and Wilde (1990) reported an isotope dilution thermal ionization mass spectrometric (IDTIMS) U–Pb upper intercept age of 2688 ± 4 Ma for multigrain zircon fractions from an outcrop sample of Teutonic Bore Formation rhyolite about 10.5 km to the southeast.

Petrographic description

The sample is a crystal-poor metarhyolite (Fig. 1), consisting of about 90% groundmass, 8–10% pseudomorphed plagioclase? phenocrysts, 1–2% quartz phenocrysts, 1% pseudomorphed biotite? phenocrysts, and trace zircon. The groundmass is composed of quartz and sericite-altered feldspar, has an average grain size of about 0.05 mm, and has undergone pervasive sericite–quartz–carbonate (phyllic) alteration and development of sericite folia. Poorly defined to tabular possible feldspar phenocrysts are completely replaced by carbonate minerals and form equant patches and lenses up to 2 mm in size. Quartz phenocrysts are fractured, slightly embayed, up to 3 mm in diameter, and exhibit moderate undulose extinction. Prismatic phenocrysts up to 1 mm long that are entirely replaced by sericite and fine-grained magnetite were probably originally biotite. Some sericite folia occur within the groundmass, although it is not clear whether this is a primary flow remnant or a tectonic fabric. The sample is not obviously deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and subhedral to euhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 6:1. Concentric zoning is ubiquitous, and a few zircons contain high-U, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11–12 June 2020, using SHRIMP-B. Nine analyses of the M257 standard were obtained during the session, of which six indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.20% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Fourteen analyses were obtained from 14 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

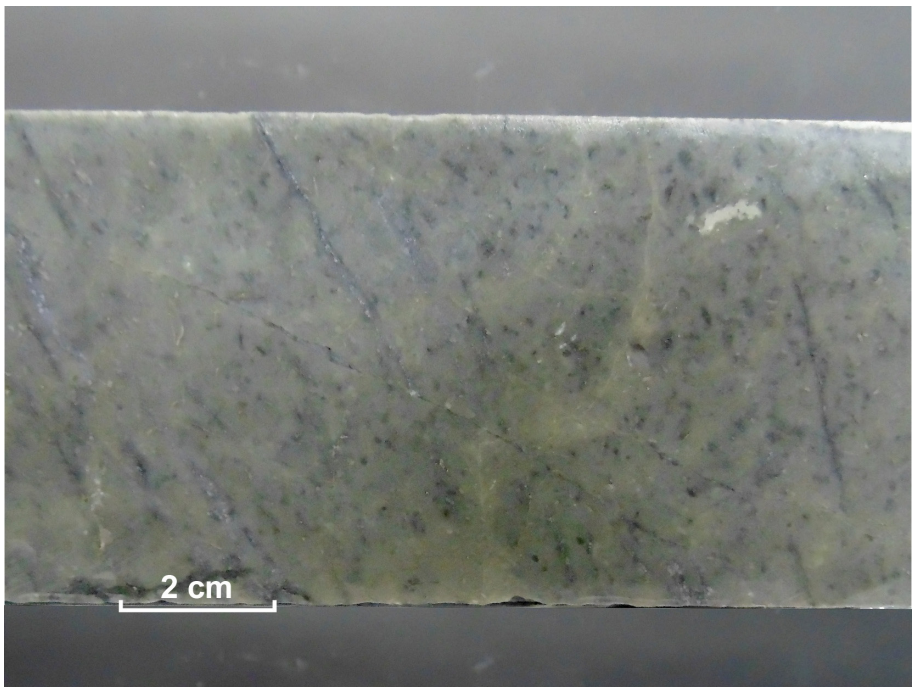

Figure 1. **Drillcore image of sample 230514: metarhyolite, Junior Hill prospect**

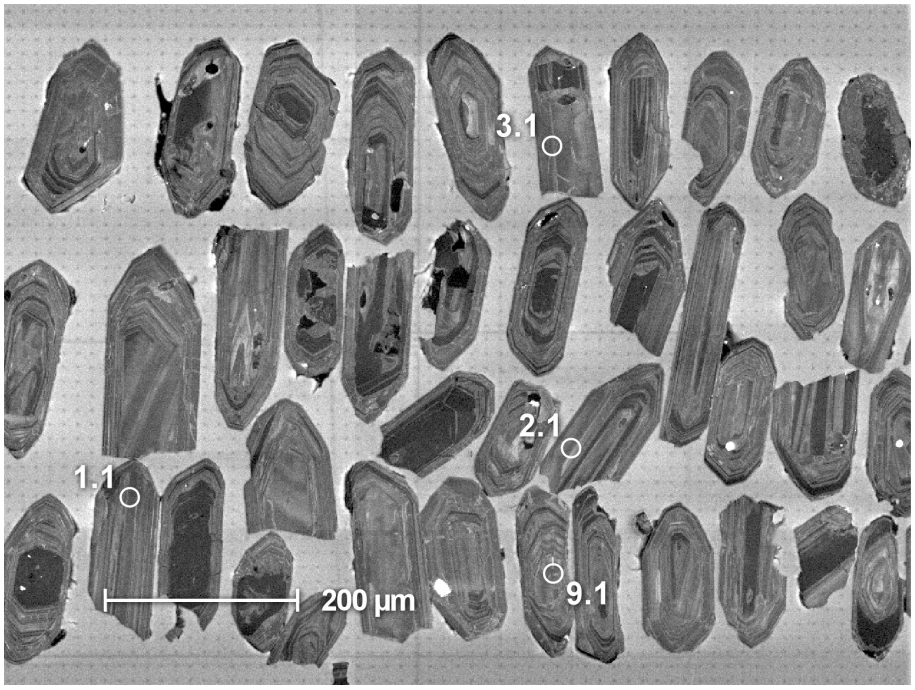

Figure 2. **Cathodoluminescence image of representative zircons from sample 230514: metarhyolite, Junior Hill prospect. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 230514: metarhyolite, Junior Hill prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	9	9.1	443	467	1.05	0.117	1.972	0.016	0.18085	0.00067	1.974	0.016	0.17982	0.00071	2642	18	2651	7	0.3
I	6	6.1	250	177	0.71	-0.027	2.016	0.018	0.18004	0.00081	2.015	0.018	0.18028	0.00082	2598	20	2655	8	2.2
I	5	5.1	194	80	0.41	0.060	2.030	0.019	0.18149	0.00090	2.031	0.019	0.18095	0.00092	2581	21	2662	8	3.0
I	1	1.1	365	478	1.31	0.042	1.974	0.017	0.18139	0.00072	1.975	0.017	0.18102	0.00073	2641	19	2662	7	0.8
I	2	2.1	294	109	0.37	0.022	1.955	0.017	0.18122	0.00076	1.956	0.017	0.18103	0.00077	2662	19	2662	7	0.0
I	14	14.1	328	266	0.81	0.039	2.024	0.017	0.18216	0.00073	2.025	0.017	0.18182	0.00074	2588	18	2670	7	3.1
P	13	13.1	621	716	1.15	0.132	2.312	0.029	0.15820	0.00059	2.315	0.029	0.15703	0.00063	2314	25	2424	7	4.5
P	10	10.1	426	199	0.47	0.067	2.106	0.017	0.17308	0.00066	2.107	0.017	0.17249	0.00068	2504	17	2582	7	3.0
P	11	11.1	364	214	0.59	0.045	2.074	0.017	0.17730	0.00070	2.075	0.017	0.17690	0.00071	2536	17	2624	7	3.4
P	7	7.1	371	204	0.55	0.157	2.074	0.017	0.17903	0.00070	2.077	0.017	0.17763	0.00076	2533	17	2631	7	3.7
P	3	3.1	398	136	0.34	0.109	2.083	0.017	0.17884	0.00071	2.085	0.017	0.17787	0.00074	2525	17	2633	7	4.1
P	4	4.1	398	195	0.49	0.056	2.009	0.016	0.17885	0.00069	2.010	0.016	0.17835	0.00070	2603	18	2638	7	1.3
Z	8	8.1	197	76	0.39	-0.015	2.954	0.029	0.11466	0.00084	2.954	0.029	0.11479	0.00085	1880	16	1877	13	-0.2
D	12	12.1	297	120	0.41	0.152	2.167	0.018	0.17319	0.00112	2.171	0.018	0.17184	0.00115	2443	17	2576	11	5.2

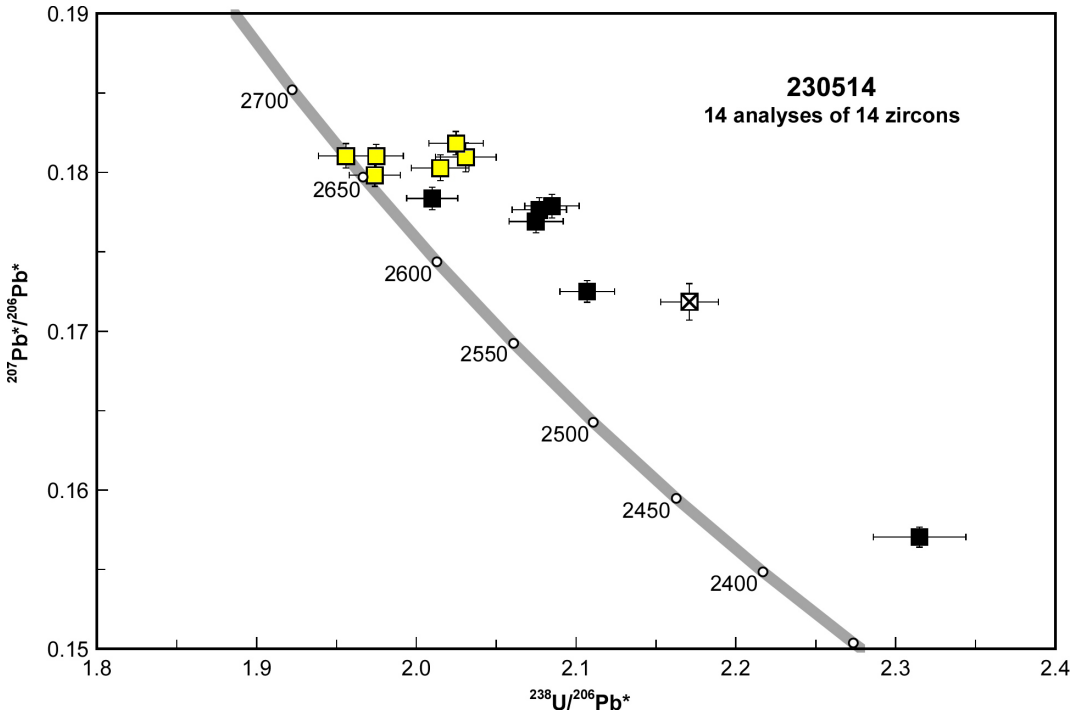

Figure 3. U–Pb analytical data for sample 230514: metarhyolite, Junior Hill prospect. Yellow squares indicate Group I (magmatic zircons); black squares indicate Group P (radiogenic-Pb loss); crossed square indicates Group D (discordance >5%). One analysis in Group Z is not shown

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). One analysis is >5% discordant. The date obtained from this analysis (Group D; Table 1) is unreliable, and is considered not to be geologically significant. The remaining 13 analyses can be divided into three groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises six analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2660 ± 6 Ma (MSWD = 0.88).

Group P comprises six analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2638–2424 Ma.

Group Z comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 1877 ± 13 Ma (1σ).

The date of 2660 ± 6 Ma for the six analyses in Group I is interpreted as the igneous crystallization age of the rhyolite. The dates of 2638–2424 Ma for the six analyses in Group P are interpreted to reflect loss of radiogenic Pb. The date of 1877 ± 13 Ma (1σ) for the single analysis in Group Z is interpreted as the age of a possible zircon contaminant introduced during sample processing.

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Recommended reference for this publication

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